MICROBIAL FOOTPRINTS: INVESTIGATING THE MICROORGANISMS INHABITING ANIMAL FEET

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BONAFIDE CERTIFICATE

Certified that this project report **"MICROBIAL FOOTPRINTS: INVESTIGATING THE MICROORGANISMS INHABITING ANIMAL FEET"** is the bonafide work of **"CAROL JACOB (URK22BT1063)"** who carried out the project for project based learning of 20BT2011 – Microbiology (III Internal) under my supervision.

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CAROL JACOB

ABSTRACT

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Dr. S.Murugan, Associate Professor

Summary:

This study delves into the often-overlooked yet intriguing world of microbial communities that thrive on the feet of various animals. Microorganisms play a vital role in shaping ecosystems, and understanding their presence on animal feet offers valuable insights into host-microbe interactions, disease transmission, and environmental impact. The study explores a diverse range of animals, shedding light on the remarkable adaptability of microbial communities to different hosts.. The findings from this investigation promise to advance our understanding of the intricate relationships between microorganisms and their animal hosts, ultimately contributing to the fields of microbiology, ecology, and animal health. This project delves into the captivating and often overlooked world of microorganisms that call animal feet home. Microbial communities on animal feet are diverse, dynamic, and play a significant role in shaping ecosystems. The study employs biotechnological and ecological methods to systematically explore these microbial footprints across a wide range of animal species.

By studying these microbial communities, the project aims to unravel the intricacies of host-microbe interactions, disease transmission dynamics, and the broader environmental impact. This research endeavour holds promise for enhancing our knowledge of the complex relationships between microorganisms and their animal hosts, with implications extending to fields such as microbiology, ecology, and animal health. Ultimately, it offers a deeper understanding of the role of microorganisms in the animal kingdom, unveiling the microbial footprints that leave an indelible mark on the natural world.

Key Words: microorganism, biotechnological, transmission, endeavor, microbiology, health, communities.

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1.INTRODUCTION

Microorganisms inhabiting animal feet play crucial roles in the overall health and behaviour of their hosts. These microscopic inhabitants aid in digestion, contribute to defence mechanisms, and even influence mating behaviours.

Understanding the microbial footprints of different animal species can offer insights into their ecology, evolution, and interactions with their environment.

This project also holds potential applications in various fields, including veterinary science, wildlife conservation, and even human health. From the tiniest insects to the mightiest mammals, animal feet play host to a diverse range of microorganisms that form unique and dynamic communities. The study of animal feet microbiomes can provide insights into the origins of certain zoonotic diseases and help us develop strategies for better disease prevention and management.

1.1 MOTIVATION AND BACKGROUND

Understanding the microbial communities on animal feet can provide insights into various health aspects. Certain microorganisms might be pathogenic and lead to diseases or infections in animals. Studying these microorganisms can help in preventing and managing diseases in both wild and domesticated animals.

The microbial communities on animal feet might influence their behavior and habitat choices. Certain microorganisms could be attracted to specific environments, affecting where animals choose to live. Studying these microbial footprints can provide valuable information about animal behavior and ecology.

1.2 AIM AND OBJECTIVES

Aim

To investigate the organisms inhabiting in animal (dogs & cats) feet.

Objectives

- To observe the variety of microorganisms, present in dog and cat feet

- To create awareness on different diseases caused by these microorganism

2. REVIEW OF LITERATURE

2.1 The Hidden World of Animal Feet:

Microbiota and microbiome, which refers, respectively, to the microorganisms and conjoint of microorganisms and genes are known to live in symbiosis with hosts, being implicated in health and disease. The advancements and cost reduction associated with high-throughput sequencing techniques have allowed expanding the knowledge of microbial communities in several species, including dogs. Throughout their body, dogs harbor distinct microbial communities according to the location (e.g., skin, ear canal, conjunctiva, respiratory tract, genitourinary tract, gut), which have been a target of study mostly in the last couple of years. Although there might be a core microbiota for different body sites, shared by dogs, it is likely influenced by intrinsic factors such as age, breed, and sex, but also by extrinsic factors such as the environment (e.g., lifestyle, urban vs rural), and diet. It starts to become clear that some medical conditions are mediated by alterations in microbiota namely dysbiosis. Moreover, understanding microbial colonization and function can be used to prevent medical conditions, for instance, modulation of gut microbiota of puppies is more effective to ensure a healthy gut than interventions in adults. (Pereira & Clemente,2021)

2.2 The Impact of Animal Feet Microbiomes

Links between host genetics and the abundance of certain taxa in the microbiome have been made in several settings and our understanding of how host genotype affects the microbiome is evolving. Mechanisms of host genetic control of the microbiome have been discussed in the context of the human gut microbiome. A genetic variant may directly cause a particular phenotype, with alterations to the microbiome being a consequence of disease. Alternatively, different genotypes may alter gene expression, which will affect the microbiome; or a genetic variant may directly alter the microbiome resulting in disease.(Bay etal., 2023)

We are learning that the host-associated animal microbiome is essential for the adaptation of the metaorganism to the environment and is therefore important for conservation biology . Woodhams etal.. evaluated the diversity and function of bacteria associated with 654 noncaptive host species to investigate the pattern of response provided by microbial communities associated with different compartments of these metaorganisms. Their results indicated that microbiomes associated with the host's external surface seem to be shaped mainly by local environmental conditions, such as daily environmental temperature and precipitation, whereas internal microbiomes seem to be regulated by host factors, including phylogeny, immune system, and diet, as well as by climate. The combination of these factors and the resulting microbiomes may be critical, as the health and behavior of a metaorganism are closely linked to the composition and function of its associated microorganisms, which may be forced to adapt whenever a change in the environment occurs, both through natural processes and through impacts caused by anthropogenic activities. These variations often increase the incidence of pathogens and, potentially, of diseases. (Raquel etal.,2021)

Microbes can increase host resilience through different mechanisms, which include nutritional input from food-conversion mechanisms, mitigation of toxic compounds, niche competition, control of pathogens, reproduction, growth, development, and rapid adaptation to environmental shifts. Microorganisms also drive host performance; shifts in the associated microbiome have been increasingly associated with animal behavior, including communication and cognitive skills and stress responses . These behavioral patterns can also, in turn, contribute to the preservation of the associated microbiome over evolutionary timescales, as observed in chimpanzees, where a pan-microbiome seems to be obtained through social behavior . Ezenwa etal.. nicely summarized correlations between animal behavior and the microbiome, discussing, for example, how bobtail squids manipulate the use of microbial symbionts (*Vibrio fischeri*), which can be used to attract prey or escape predators. In addition, the search performed by iguanas for the right recipe for gut microbial composition and the key role of insect nymphs' in the acquisition of inherited microbiomes that guarantee their survival are also good examples correlating the microbiome and their host's behavior . (Raquel etal.,2021)

The phenotypic reaction of a metaorganism to environmental shifts or disease can be considered as a combined genomic, transcriptomic, and proteomic response of the host and its associated microbiome. Different organisms are facing unprecedented pressures globally. Alarming declines in the abundance and diversity of corals, frogs, insects, bats, birds, and other organisms are occurring across most continents, driven by the cumulative impacts of local stressors and rapidly changing climate. In most cases, these impacts also trigger dysbiotic processes that affect key members of the host-associated microbiomes. Dysbiosis, or imbalance of the healthy microbiome, can be defined as the process of a decrease in abundance or loss of key symbiotic relationships within the metaorganism . Similarly to some chronic diseases that occur in humans, this process has also been proposed for several other organisms; dysbiotic processes frequently result in infection by opportunistic pathogens and polymicrobial infections in corals, frogs, and other organisms. Likewise, dysbiotic processes can impact animal productivity and therefore threaten food production. Addressing the major questions of dysbiosis and attempting to improve animal health require a better understanding of the basic concepts of microbiome diversity and distribution, disease and pathogenicity, symbiotic mechanisms and interactions, and ways to control these microbial populations. Here we explore some examples to summarize the current overall knowledge of the diversity of wild and domesticated animal-associated microbiomes, their potential beneficial roles, and the state of the art of microbial-based therapies to promote animal health. (Raquel etal.,2021)

2.3 Microbiomes and Animal Health

The massive decrease in sequencing costs associated with the generalization of high-throughput or next-generation sequencing (NGS) techniques has enabled unprecedented advances in microbiome studies spanning throughout the life sciences fields. The strong bond between public health and the economy has been propelling the interest in

microbiome research, which is deemed to hold a huge applicative potential under the One Health strategy and other similar initiatives. Most of such studies addressing health and animal production have been mostly focused on gut microbiota, which is justified by the crucial role of these microorganisms in nutrition, fitness, and performance traits .It is generally expected that advancing knowledge of the ruminant microbiome bears a huge potential in terms of boosting animal production and health while lessening environmental. This promise seems of utmost importance when considering forecasts predicting an almost two-fold increase in the current production and consumption of meat in 30 years from now, with changes in dietary habits in developing countries—on top of human population growth—which will boost the demand for dairy products. (Forcina etal., 2022)

Some terms will be used extensively in this review and, for the benefit of the readers, their definition is provided as follows. The community of microorganisms inhabiting a given environment is referred to as a microbiota, while the term microbiome is used to indicate microbiota's collective genomes. On the other hand, the concept of metagenomics, first defined as "the direct genetic analysis of genomes contained in an environmental sample", has been elaborated further as the "study of the structure and function of entire nucleotide sequences isolated and analyzed from all the organisms (typically microbes) in a bulk sample”. (Forcina etal., 2022)

However, the fast-increasing body of research produced in the wake of the above-mentioned compelling socioeconomic reasons has yet to cover much ground. Among the main shortfalls of this kind of study is the almost exclusive focus on cosmopolitan and highly selected breeds, which lacks representativeness both in terms of diversity and functionality, as the most promising knowledge may come from locally adapted native breeds. The role of microorganisms in the resilience and performance of livestock species is of paramount interest for their potential commercial value, especially in a time of rampant global change. (Forcina etal., 2022)

Another critical factor is the uneven attention being paid to bacteria, which include taxa that cause significant economic losses in addition to being a serious hazard to public health, e.g.,. On the other hand, the other microorganisms, such as fungi, archaea, protozoa, and viruses, have received far less attention. Yet another weakness is that the classical metabarcoding approach is still largely used, as opposed to increasingly feasible and affordable shotgun approaches that are now available (although only for a low number of samples) for a more precise and extensive characterization of microbial communities. (Forcina etal., 2022)

2.4 Animal-Microbe Coevolution

Humans have evolved with their pathogens, parasites, and commensals. This process of coevolution has shaped human physiology as well as the biology of the microbiota. The human gut microbiome contributes to pathogen defense, metabolic health, and the

regulation of the immune system. The specificity of the adaptive immune system is a consequence of Darwinian processes of variation and selection. Trade-offs between virulence and transmissibility influence the progression of microbial infections in individuals and populations. These attributes of the pathogen can evolve rapidly, with important consequences for therapeutic strategies. Public health measures, vaccination, and antibiotics have greatly reduced the threat from infectious disease in high-income countries, but infection is still a major cause of morbidity and mortality in low-income countries. The development of antimicrobial resistance in pathogens can be understood in terms of an evolutionary arms race, providing insights into how to reduce this risk by adjusting treatment regimens. (Gluckman etal. 2016)

2.5 Diversity of Animal Feet Microbiomes

Bacteria are ubiquitous in the environment and constitute the majority of life on Earth. They inhabit not only abiotic environments but also bodies of other living organisms. Being present in virtually all tissues that are in contact with the external environment, the diversity of such bacteria together with their underlying genotypic variation forms what is called a microbiome: a unique, taxon-specific and highly plastic collection of not only various bacterial but also fungal, protozoan and viral species that interact synergistically with an organism's physiology and biochemistry and contribute to genetic variation underlying host's traits (Hird 2017, Hicks etal.. 2018).

There is ample evidence of the impact microbiome diversity has on individual fitness. A huge majority of available published results elaborates on the importance of microorganisms in digestion and nutrients assimilation (Hird 2017). Symbiotic bacteria can also affect and modulate components of individual immune response, can affect the general metabolic homeostasis of an organism and finally, they can also modulate individual behaviours through links that exist between organs hosting symbiotic bacteria (e.g.gut) and the brain (Toivanen etal.. 2001, Mazmanian etal.2005)

These physiology and condition links often manifest themselves as differences in microbiotic diversity existing between sexes, age classes or developmental stages of individuals within populations, especially if different classes of individuals in populations engage in different types of behaviours or occupy different ecological niches (Spor etal.. 2011). For instance, sex differences likely begin to play a role already in the prenatal period in mammals, when developing embryos are exposed to compounds secreted by adult female microbiome of their mothers (Jašarević etal. 2016)

2.6 The Canine Foot Microbiome

Advancements made in high-throughput sequencing techniques have allowed sophisticated analysis and unprecedented knowledge of microbial communities. Techniques such as amplicon and shotgun sequencing coupled to advanced bioinformatics have been employed to unravel the composition and diversity of communities of dogs in healthy and disease states. Thus, enabling a more integrative understanding of the physiological and pathological processes occurring, as well as the influence exerted by host-microbe interactions. As a result, one can search for more accurate therapeutic interventions sustained in a holistic view of dogs' phenotype and their environmental determinants. (Ana & Alfonso, 2021)

The acknowledged preponderant role of microbial communities of the gastrointestinal tract (GIT) on dog's health explains that most records focus microbiome of the gut. However, the composition of the microbial communities in other body sites (e.g., skin, ear, eye, and other internal structures) have also been a recent target of study. In healthy dogs, the composition and abundance of microorganisms vary according to the body site , which is in turn, related to the functionality of the communities.(Ana & Alfonso, 2021)

2.7 The Feline Foot Microbiome

As with other species, the skin microbiome of cats has been assessed over the past few years utilizing modern technologies. This has resulted in the identification of many more bacterial and fungal organisms compared with what had been recorded historically on the skin in various states of health and disease using culture-based studies. This information is expanding the knowledge of how microbial communities are impacted by various changes in the skin health of cats. More specifically, how these microbial communities change in the face of health and disease, and how various therapeutic interventions affect the cutaneous microbiome, lends a greater understanding of disease pathogenesis and provides a growing area of research for correcting dysbiosis and improving feline skin health.(Older etal)

Despite the inherent complexities of the gut microbiome and its interactions with hostsystems, microbiome data is advancing our collective understanding of health and disease. For example, a number of chronic diseases afflicting human populations have significantly altered gut microbiome compositions when compared to the microbiomes from healthy cohorts. These conditions include obesity, diabetes, irritable bowel syndrome (IBS), peri-odontitis, and skin diseases, among other metabolic and immune-related conditions. Likewise domestic cats have health conditions that are also associated with imbalances (dysbiosis) in the composition of gut bacteria, such as Inflammatory Bowel Disease (IBD),lymphoma, diabetes, periodontal disease, and atopic dermatitis [7–

11].Because individuals have distinctly personal microbiome signatures, aggregating microbiome data from many individuals within a natural population yields a dataset with significant compositional heterogeneity. As a result, simply characterizing the composition of a healthy microbiome is both a major challenge and a necessary first step towards identifying disease state-associated imbalances [12–14]. This is important for not only elucidating microbial signatures of disease states, but also understanding how the presence and/or abundance of specific taxa might be manipulated to drive the microbiome composition towards that of the disease-free state [12].As evidenced by human studies, it

is difficult to define a singular healthy host microbiome due to vast differences in the diets, lifestyles, geographic regions, and other factors that influence the composition of microbial taxa in individuals [15–17]. To date, studies that investigate associations between the microbiome and disease typically define 'healthy' as the absence of disease, but this does not necessarily mean that these individuals are 'healthy' [18]. Therefore, it is essential to build a comprehensive dataset of the microbiome composition of healthy individuals to establish the expected ranges for the structure of these communities within a healthy population. Such a database can then be rigorously interrogated to, for example, understand therapeutic targets. By reducing variation through a carefully defined healthy reference set, we will have more power to detect signals associated with disease states.(Ganz etal., 2022)

2.8 Behavior and Communication

Cellular communications play pivotal roles in multi-cellular species, but they do so also in uni-cellular species. Moreover, cells communicate with each other not only within the same individual, but also with cells in other individuals belonging to the same or other species. These communications occur between two unicellular species, two multicellular species, or between unicellular and multicellular species. The molecular mechanisms involved exhibit diversity and specificity, but they share common basic features, which allow common pathways of communication between different species, often phylogenetically very distant. These interactions are possible by the high degree of conservation of the basic molecular mechanisms of interaction of many ligand–receptor pairs in evolutionary remote species. These inter-species cellular communications played crucial roles during Evolution and must have been positively selected, particularly when collectively beneficial in hostile environments. It is likely that communications between cells did not arise after their emergence, but were part of the very nature of the first cells. Synchronization of populations of non-living protocells through chemical communications may have been a mandatory step towards their emergence as populations of living cells and explain the large commonality of cell communication mechanisms among microorganisms, plants, and animals. (Combarnous,2020)

2.9 Zoonotic Potential

As people and a wide variety of animal species have increasingly close contact in diverse settings, guidance on preventing zoonotic diseases, caused by pathogens that spread between animals and people, is urgently needed. According to data (Annual Report of Animal Contact Outbreaks, 2021; Reptiles and Amphibians, 2022; Salmonella from Small Mammals, 2021; US Outbreaks of Zoonotic Diseases Spread between Animals & People, 2021) from the Centers for Disease Control and Prevention (CDC), three major groups of animals have repeatedly been associated with local, regional, and national outbreaks of zoonotic diseases in people in the United States: rodents, backyard poultry, and reptiles. This Compendium presents information on these and other non-traditional pet (NTP)*

animal species associated with a high risk of zoonotic disease transmission in any setting. Other animal species covered in this Compendium include non-rodent small mammals (e.g., hedgehogs, ferrets), amphibians (e.g., African dwarf frogs), and other aquatic species (e.g., fish, coral) that are less frequently linked to illness or outbreaks, but nonetheless pose a risk of zoonotic disease transmission. (Varela etal., 2022)

Many zoonotic disease exposures occur at home through direct or indirect contact with pets, agricultural animals, or feeder animals. Zoonotic diseases also affect workers employed in various segments of the pet industry, including animal breeders, pet store and other retail employees, and pet importers and distributors, as well as volunteers working closely with animals. This Compendium will provide guidance on addressing the zoonotic disease risks related to NTPs that are specific to these groups and settings, which may be different from those for other settings, populations, and animals. (Varela etal., 2022)

Zoonotic pathogens associated with NTP species and other animals can be transmitted by direct or indirect animal contact. Direct transmission of zoonotic pathogens can occur when people pet, touch, or kiss animals via accidental ingestion of feces from contaminated fur, feathers, scales, or spines; contaminated saliva from feeding an animal by hand or being licked by animals; or contact with other body fluids (Daly etal., 2017).

Direct transmission of zoonotic pathogens can also occur via animal bites and scratches. Though rare, rabies, tetanus, and other highly pathogenic infections such as infections with Streptobacillus moniliformis (rat bite fever), Francisella tularensis (tularemia), or Pasteurella can be associated with NTP bite and scratch injuries, depending on the NTP species involved (Daly etal., 2017).

Some NTP species such as venomous arthropods (e.g., scorpions and centipedes), arachnids (e.g., spiders), reptiles, amphibians, and fish can cause envenomation through stings, bites, or other contact. NTPs may also carry parasites (e.g., ticks, fleas) that may transmit vector-borne diseases to people (Daly etal., 2017)

2.10 Future Directions in Microbial Footprint Research

Two notions broadly accepted in developed western societies have made animal care guidelines inevitable. These are that domestic animals are sentient and that humans are responsible to ensure the proper care of domestic animals. Despite these common views, people have differing moral understandings of the human-animal relationship, and there are sharp divisions over how these views should be applied to domestic animal care. Animal care guidelines have been developed by different nations at several organizational levels to represent a compromise that is acceptable to most people. These organizational levels include individual poultry companies, national poultry associations, individual customers of the poultry industry, national associations of customer companies, national governments, and international organizations. Animal care guideline development has typically included input from producers and scientists and, depending on the sponsoring organization, animal advocates and government representatives as well. Animal advocacy groups have also sought to influence domestic animal care by campaigning against animal production practices or by offering their preferred guidelines for producers to adopt in the hope that the endorsement of the welfare group would add value to the product. Originally, animal care guidelines were only recommended, with little or no requirement for compliance. In recent years, the need for retail companies to assure certain welfare standards has led to animal welfare auditing of production facilities. Animal care guidelines primarily have sought to establish standards for handling and husbandry in existing production systems. Future guidelines may put increasing emphasis on adoption of alternative management practices or housing systems. International animal care guidelines are being developed on 2 levels (i.e., among national governments to create a common standard for trade in animal products and within international retail companies to create company-wide animal care standards). These initiatives should tend to unify farm animal care standards worldwide but perhaps at a level some nations might consider lower than preferable. (Webster,2007)

The relationship between microbial community and host has profound effects on the health of animals. A balanced gastrointestinal (GI) microbial population provides nutritional and metabolic benefits to its host, regulates the immune system and various signalling molecules, protects the intestine from pathogen invasion, and promotes a healthy intestinal structure and an optimal intestinal function. With the fast development of next-generation sequencing, molecular techniques have become standard tools for microbiota research, having been used to demonstrate the complex intestinal ecosystem. Similarly, to other mammals, the vast majority of GI microbiota in cats (over 99%) is composed of the predominant bacterial phyla Firmicutes, Bacteroidetes, Actinobacteria, and Proteobacteria. Many nutritional and clinical studies have shown that cats' microbiota can be affected by several different factors including body condition, age, diet, and inflammatory diseases. All these factors have different size effects, and some of these may be very minor, and it is currently unknown how important these are. Further research is needed to determine the functional variations in the microbiome in disease states and in response to environmental and/or dietary modulations. Additionally, further studies are also needed to explain the intricate relationship between GI microbiota and the genetics and immunity of its host. This review summarizes past and present knowledge of the feline GI microbiota and looks into the future possibilities and challenges of the field. (Yang etal., 2020)

3. METHODOLOGY

3.1 MATERIALS AND EQUIPMENT:

- Nutrient agar
- Agar agar
- Distilled water
- A heat source
- A balance for measuring the agar and nutrient agar
- An autoclave (for sterilization)
- Petri dishes

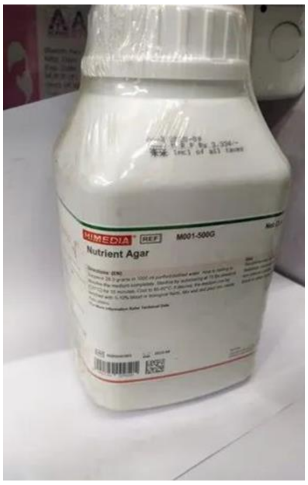

Fig 1. Nutrient Agar

Fig 2.Agar Agar

3.2 Procedure for bacterial media preparation

In the laboratory experiment, 2.80 grams of nutrient agar and 2.53 grams of agar agar were precisely measured using a weighing balance. Subsequently, 100 ml of distilled water was added to the container containing the agar and nutrient agar. The mixture was thoroughly stirred to ensure the even distribution of the agar components in the water and then subjected to sterilization. After the sterilized mixture had cooled down, it was carefully poured into sterile Petri dishes. The poured medium was left to cool and solidify in the Petri dishes for approximately 15 minutes at room temperature. Once solidified, the sterilized Petri dishes containing the agar medium were stored in a cool, dry place until they were needed for bacterial culture. This meticulous process ensured the preparation of a suitable and sterile agar medium for cultivating bacteria in the laboratory setting.

Fig 3: Nutrient Agar measurement

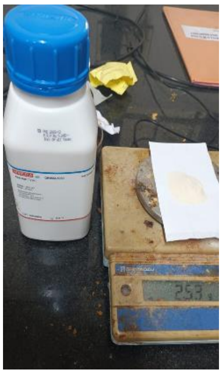

Fig 4: Agar Agar measurement

3.3 Material used for fungal media:

- Potato Dextrose Agar (PDA)
- Agar agar
- Distilled water
- A heat source
- A balance for measuring the agar and nutrient agar
- An autoclave (for sterilization)
- Petri dishes

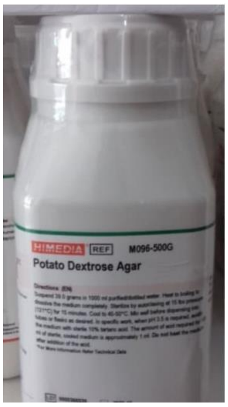

Fig 5: Potato Dextrose Agar

Fig 6: Agar agar

3.4 Procedure for fungal media preparation

In the laboratory procedure, 3.93 grams of potato dextrose agar and 2.53 grams of agar agar were accurately measured using a balance. Subsequently, 100 ml of distilled water was added to the container containing the agar and potato dextrose agar. The mixture was stirred thoroughly to ensure the even distribution of the agar components in the water and then subjected to sterilization. Once the sterilized mixture had cooled down, it was carefully poured into sterile Petri dishes. The poured medium was left to cool and solidify in the Petri dishes for approximately 15 minutes at room temperature. After solidifying, the sterilized Petri dishes containing the agar medium were stored in a cool, dry place until they were needed for fungal culture. This meticulous process ensured the preparation of a suitable and sterile agar medium for cultivating fungi in the laboratory setting.

Fig 7: PDA measurement

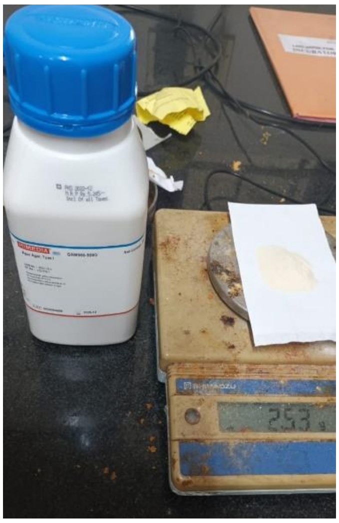

Fig 8: Agar agar measurement

Fig 9: Pouring the media

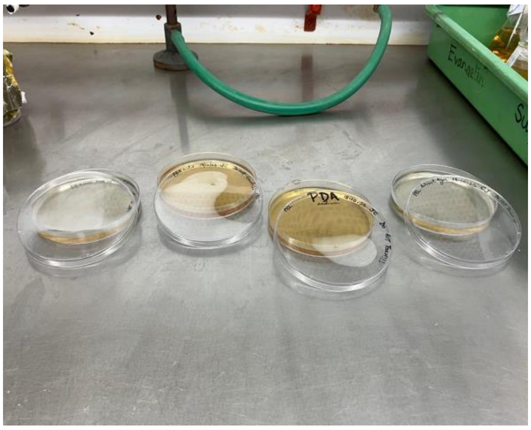

Fig 10: Labelling the petri dishes

3.5 procedure for making saline solution

In the experiment, a beaker was used to combine 1 gram, roughly equivalent to a teaspoon, of sodium chloride with approximately 80 milliliters of distilled water. The sodium chloride was carefully added to the beaker, followed by the addition of the water. The mixture was stirred diligently until all the salt particles had completely dissolved in the water, resulting in a clear solution. Additionally, three cotton buds were placed in a separate container and sealed for sterilization, likely to be used in the experiment. This process ensured the creation of a saline solution and sterilized cotton buds, both crucial elements for scientific procedures and experiments.

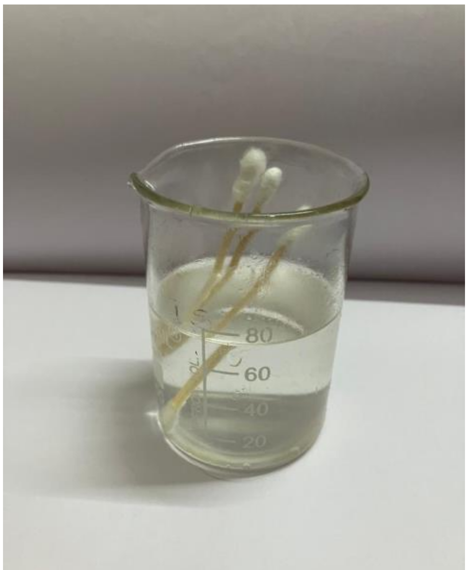

Fig 11: Saline solution

3.6 Sampling

Fig 12: Oskie's Pawprint Sampling

- With the cotton bud in the saline solution, we rubbed the bud on the paw of the dog.
- Stored in a plastic bag for further inoculation in the laboratory

Fig 13: Cookie’s Pawprint Sampling

- With the cotton bud in the saline solution, we rubbed the bud on the paw of the cat.
- Stored in a plastic bag for further inoculation in the laboratory

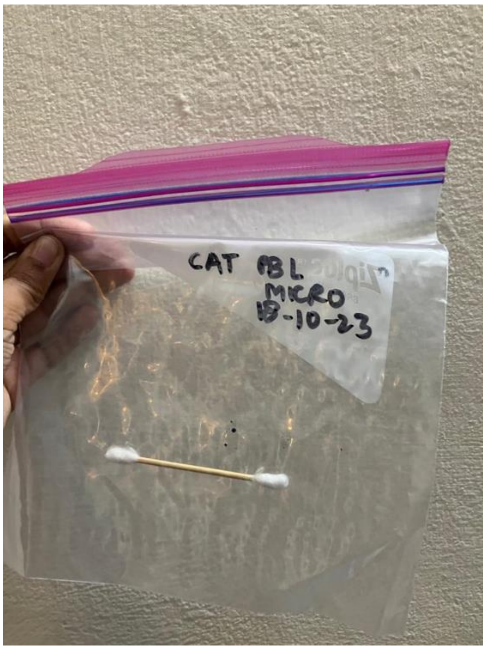

Fig 14: Cat Paw Swab

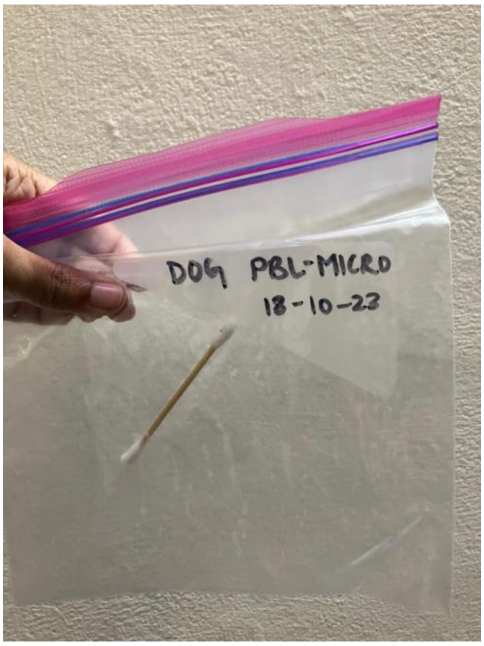

Fig 15: Dog Paw Swab

4.RESULTS

4.1 Results Dog Swab

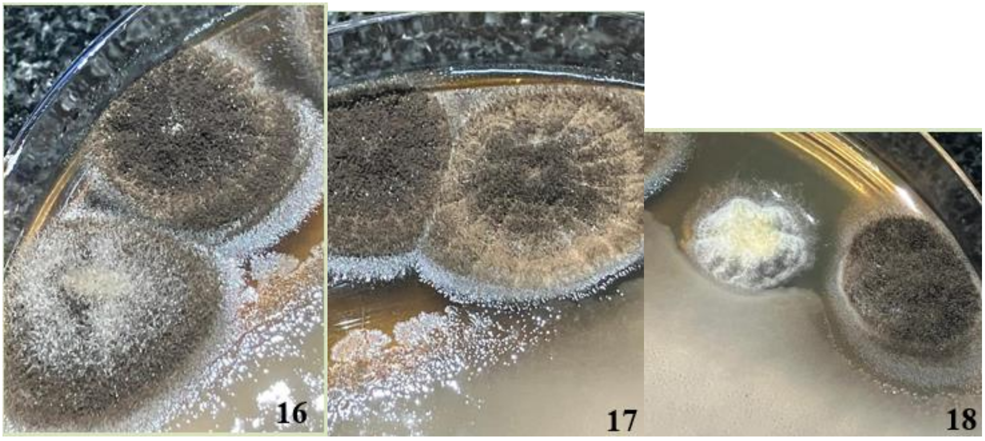

Fig 16, 17 and 18 Close Up Of The Fungus

Fig 19: Fungus Grown on Dog Swab Sample

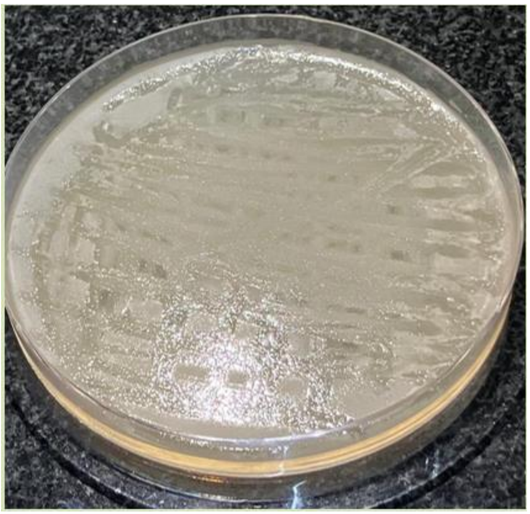

Fig 20: Bacteria in Dog Sample

The higher occurrence of fungi on dog paws raised significant questions concerning canine health. Fungal infections, particularly yeast infections, were found to lead to discomfort, itchiness, and dermatitis in dogs. These findings underscored the importance of regular paw hygiene and grooming practices for dogs. Pet owners were encouraged to clean and inspect their dog's paws routinely, as this practice could significantly reduce the risk of fungal overgrowth and associated health issues. Additionally, maintaining a clean living environment for dogs, including their bedding and play areas, could help prevent the proliferation of these fungi.

4.2 Results in Cat Swab

Fig 21: Close Up Of the bacteria

Fig 22: Bacteria Grown On cat sample

Fig 23: Fungus In Cat Sample

Cats are more meticulous groomers than dogs, and they use their tongues to clean themselves thoroughly. This grooming behavior can help in removing fungal spores from their paws, reducing the likelihood of fungal growth. On the other hand, bacteria are naturally present on the skin and are constantly introduced through various environmental sources, leading to a higher bacterial count.

The pH level and moisture content of a cat's skin can create an environment that is more conducive to the growth of bacteria rather than fungi. Bacteria often thrive in slightly acidic environments, which might be the case with cat skin. Additionally, the moisture levels on the paw surface can also influence microbial growth, with bacteria being more adaptable to a slightly moist environment than fungi.

5. DISCUSSION

The study's results unveiled a striking contrast in the microbial compositions found on the feet of dogs and cats. Remarkably, researchers observed a higher prevalence of fungal species on the paws of dogs, while cat feet exhibited a greater abundance of bacterial communities. These findings pointed to distinct microbial footprints associated with each species, providing valuable insights into potential implications for animal health, zoonotic risks, and avenues for preventive measures.

Conversely, the prevalence of bacteria on cat feet highlighted the need for vigilant zoonotic disease management. Cats were found to carry bacteria on their paws, posing a potential risk to their human companions, especially for individuals with compromised immune systems. Regular hand washing and avoiding close contact with cat paws were deemed essential. Furthermore, cat owners were advised to ensure that their pets were up-to-date on vaccinations and received routine veterinary care to minimize the risk of bacterial transmission.

To raise awareness of the intricate relationship between animal foot health and human well-being, educational initiatives were deemed imperative. Public campaigns, workshops, and online resources could be developed to inform pet owners about the importance of regular paw care and hygiene practices. Veterinarians and pet care providers played a pivotal role by offering guidance on maintaining healthy paws and discussing zoonotic risks with pet owners during routine check-ups.

6. CONCLUSION

The exploration into the intricate realm of "Microbial Footprints: Investigating the Microorganisms Inhabiting Animal Feet" has shed light on the fascinating interactions between animals and their foot microbiomes. The study revealed a significant contrast between dogs and cats, with dogs harbouring a higher abundance of fungi and cats carrying a greater bacterial load on their feet. These findings highlighted the complexity and distinctiveness of the microbiological ecosystems present on the paws of beloved animal companions.

Conversely, the higher presence of bacteria on cat feet emphasized the need for careful zoonotic disease management. Regular handwashing and cautious contact with cat paws were deemed essential, especially for individuals with compromised immune systems. Raising awareness about the potential bacterial transmission from cats to humans proved vital for public health.

The project did not conclude with these findings; rather, it served as a foundation for broader implications. Disseminating knowledge on the delicate balance between animal foot health and human well-being became essential. Education and awareness campaigns, guided by veterinarians and pet care providers, empowered pet owners to maintain healthy paws for their animals and stay vigilant about zoonotic risks. Such awareness efforts fostered a culture of responsible pet ownership, strengthening the bonds between humans and their animal companions.

In essence, the project underscored the interconnectedness of all life forms on Earth. Understanding and respecting the microbial footprints of animals, such as dogs and cats, enriched humanity's understanding of the natural world and empowered people to coexist harmoniously with their furry friends. As society moved forward, the exploration of intricate relationships between the animal kingdom and microbial life continued, aiming to preserve and protect the health and well-being of both animals and humans alike.

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